

b-secure — Database Design Index

Version: 1.0.0

This folder contains the complete database design documentation for b-secure, split by service.

Folder Structure

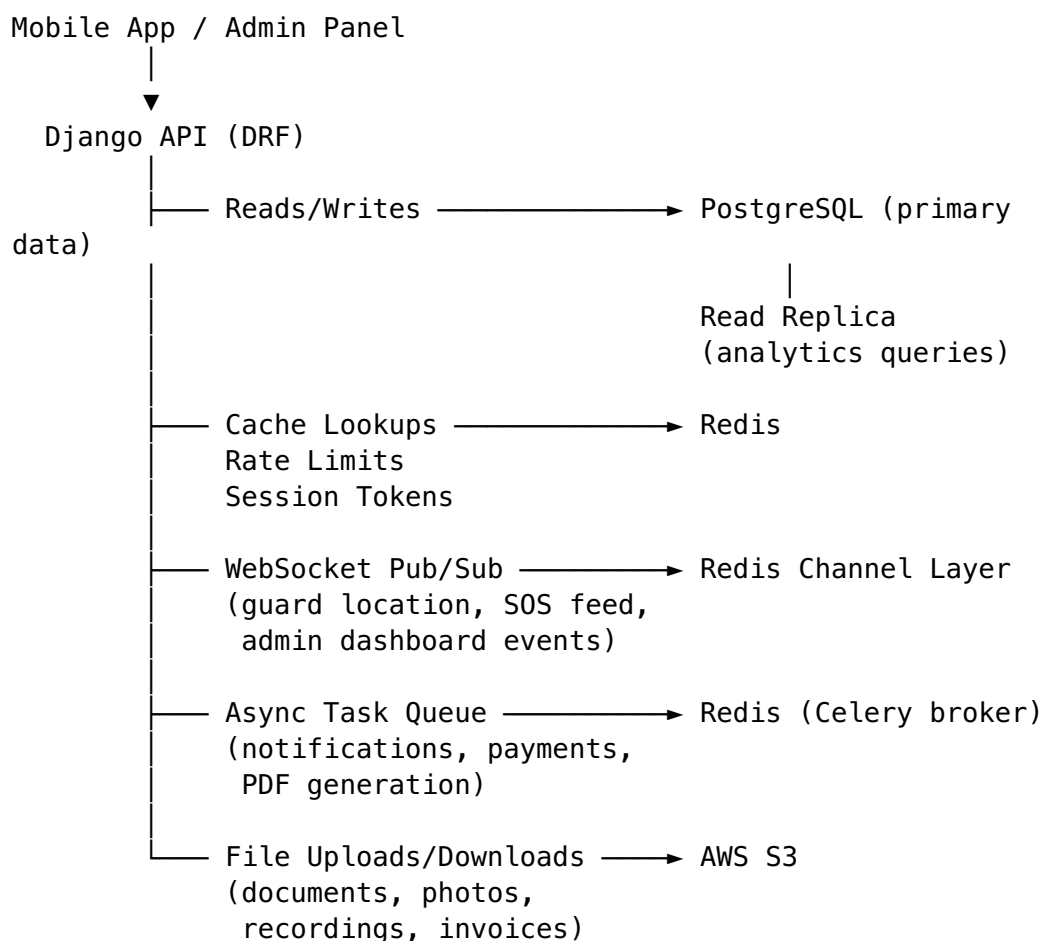
database/	
— README.md	← This file (index + overview)
— postgres/	← Primary relational database
— schema.md	← All tables, columns, data types,
constraints	
— indexes.md	← Index strategy, GiST spatial
indexes, partial indexes	
— migrations.md	← Migration workflow, PostGIS
setup, data migrations	
— queries.md	← Common queries, geospatial
queries, analytics	
— redis/	← In-memory data store (multi-
purpose)	
— design.md	← Key naming, TTLs, data
structures, all use cases	
— celery_broker.md	← Celery queue design, task
payloads, monitoring	
— aws_s3/	
— design.md	← Bucket layout, IAM policies,
lifecycle rules, access patterns	

Service Responsibilities

Service	Role	Data Stored
PostgreSQL + PostGIS	Primary source of truth	All business data: users, guards,

Service	Role	Data Stored
		bookings, payments, SOS, reviews, analytics
Redis	Cache + real-time backbone	Sessions, rate limits, Django Channels pub/sub, Celery task queues, guard proximity cache
AWS S3	Object storage	Guard documents, profile photos, session recordings, invoice PDFs, static assets

High-Level Data Flow



Database Stats Estimates (Year 1 projections)

Table	Estimated Rows	Growth Rate	Notes
users_profile	100,000	+5,000/month	Users + guards
guards_profile	10,000	+500/month	10% of users are guards
bookings_booking	500,000	+50,000/month	Core transaction table
tracking_location_snapshot	300,000,000	~1M/day	High volume — archived after 90 days
payments_transaction	1,500,000	+150,000/month	3 transactions per booking avg
notifications_log	5,000,000	+500,000/month	~10 notifs per booking
sos_sosalert	5,000	~100/month	Low volume, high priority

Storage estimates (Year 1): - PostgreSQL: ~50 GB data + ~20 GB indexes - Redis: ~2 GB peak (mostly ephemeral) - S3: ~500 GB (documents + recordings + invoices)